

Principles of Database

Report On MEDICAL BILL

1. Rationale:

- As this is generic software, it can be used by a wide variety of outlets (Retailers and Wholesaler's) to automate the process of manually maintaining the records related to the subject of maintaining the stock and cash flows.

2. Aim/Benefits of the Project: We are creating database of medical bill in a Medical Shop for keeping records of different medicines as well as of customers also. Benefits of project are to:

- To provide storage structure for efficient querying process.
- To restrict unauthorized access to data.
- To provide multiple interfaces to different classes to users.
- To represent complex relationship among data.

3. Course Outcomes Addressed:

- Use fundamental concepts of database in a Database System.
- Develop relational databases using Entity Relationship modelling approach.
- Choose relevant data model to solve a problem.
- Apply constraints for data integrity in Relational Database.
- Apply data Normalization and Denormalization techniques on database.

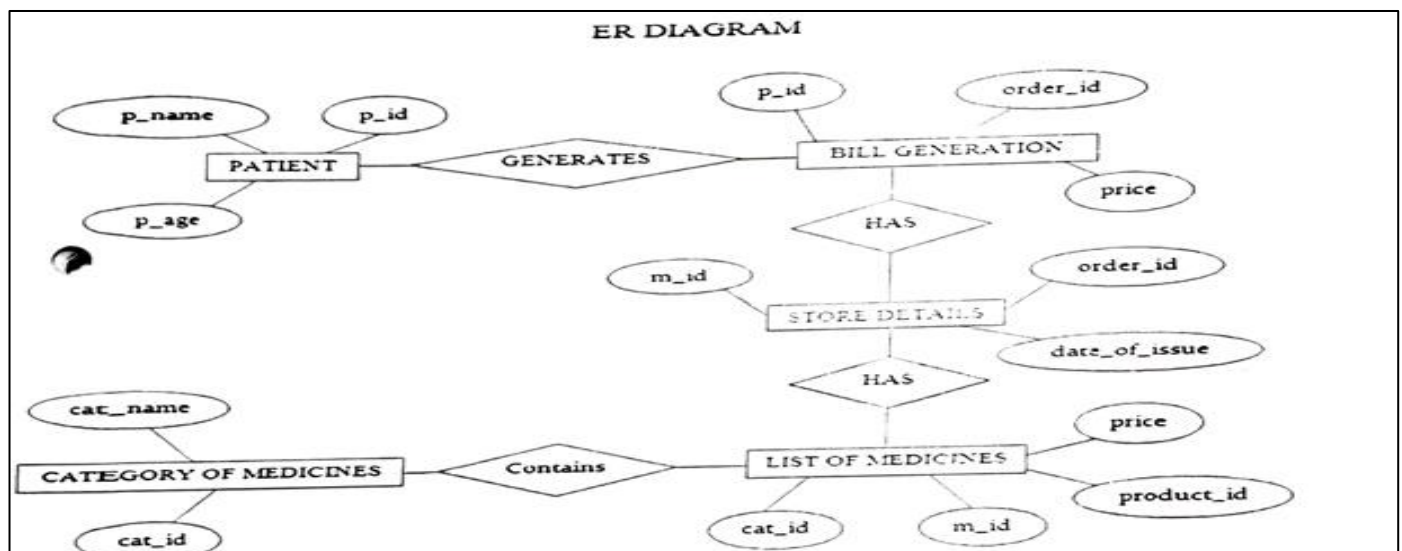
4. Literature Review:

❖ Theory of database:

- This project is basically updating the manual chemist inventory system to automated inventory system, so that the organization can manage the record in efficient and organized form.
- The main goal of this project is to maintain the records of purchase. Sales and stock details with cash transaction maintenance.
- This is helpful them to maintain day to day transactions.

5. Actual Methodology Followed:

- Creation of ER Diagram.



- Indexing the entities together, enforcing integrity constraints into the database.

All Tables		Patient		Field Name		Data Type	Description
Patient : Table			p_id		Number		
Generation : Table			p_name		Text		
Bill_Generation : Table			p_age		Number		
Store_Details : Table							
List_of_medicines : Table							
Category_of_medicines : Table							

Field Properties	
General	Lookup
Field Size	Long Integer
Format	
Decimal Places	Auto
Input Mask	
Caption	
Default Value	
Validation Rule	
Validation Text	
Required	No
Indexed	Yes (No Duplicates)
Smart Tags	
Text Align	General

The screenshot displays the Microsoft Access interface. On the left, the 'All Tables' pane shows a list of tables organized by category:

- Patient**
 - Patient Table
- Bill Generation**
 - Bill Generation Table
- Store Details**
 - Store Details Table
- List of medicines**
 - List of medicines Table
- Category of medicines**
 - Category of medicines Table

The main area shows the 'Field Properties' task pane for the 'order_id' field in the 'Bill Generation' table. The field's data type is 'Number'. The 'Field Properties' pane is open, showing the following settings:

- Field Size:** Long Integer
- Format:** (empty)
- Decimal Places:** Auto
- Input Mask:** (empty)
- Caption:** (empty)
- Default Value:** (empty)
- Validation Rule:** (empty)
- Validation Text:** (empty)
- Required:** No
- Indexed:** Yes (No Duplicates)
- Smart Tags:** (empty)
- Text Align:** General

Field Name	Data Type	Description
store_id	Number	
order_id	Number	
m_id	Date/Time	
date_of_issue	Date/Time	

Field Properties

General | Lookup

Field Size: Long Integer

Format:

Decimal Places: Auto

Input Mask:

Caption:

Default Value:

Validation Rule:

Validation Text:

Required: No

Indexed: Yes (No Duplicates)

Smart Tags:

Text Align: General

Tables

Tables:

- Patient : Table
- Bill_Generation : Table
- Store_Details : Table
- List_of_medicines : Table
- Category_of_medicines : Table

List_of_medicines

Field Name	Data Type	Description
m_id	Number	
m_name	Text	
cat_id	Number	

Field Properties

General

Field Size: Long Integer

Format:

Decimal Places: Auto

Input Mask:

Caption:

Default Value:

Validation Rule:

Validation Text:

Required: No

Indexed: Yes (No Duplicates)

Smart Tags:

Text Align: General

All Tables

Tables:

- Patient : Table
- Bill_Generation : Table
- Store_Details : Table
- List_of_medicines : Table
- Category_of_medicines : Table

Category_of_medicines

Field Name	Data Type	Description
cat_id	Number	
cat_name	Text	

Field Properties

General

Field Size: Long Integer

Format:

Decimal Places: Auto

Input Mask:

Caption:

Default Value:

Validation Rule:

Validation Text:

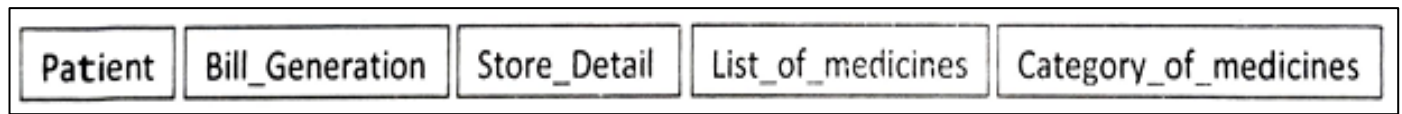
Required: No

Indexed: Yes (No Duplicates)

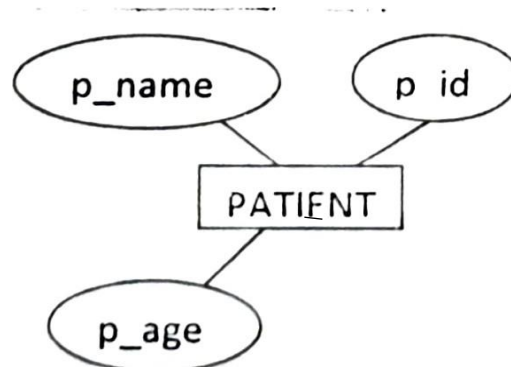
Smart Tags:

Text Align: General

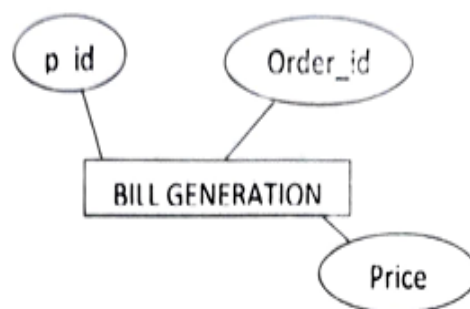
- To normalize the database.
- This is an Entity Relationship Diagram which is a graphical representation of an information system that depicts relationships between objects, people, concepts or events within the system.
- This is an Entity Relationship Diagram representing Information about Medical Bill System. It starts with the graphical representation of entities such as Patient, Bill Generation, Store details, List of medicines and Category of medicines.



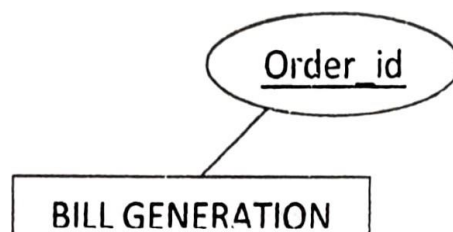
- Each Entity has a set of attributes.
- When Patient buys medicines a Bill gets generated. This Entity consists different attributes such as p_id (Patient_ID), p_name (Patient Name) and p_age (Patient Age).



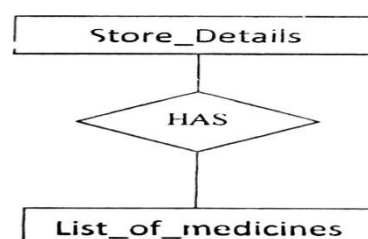
- Bill consists of store information which has date of Issue of the medicine, P ID, Price and Order ID.



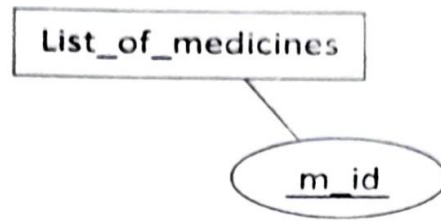
- In the Entity "Bill Generation", Order ID is the primary key as it uniquely identifies other attributes of the same entity.



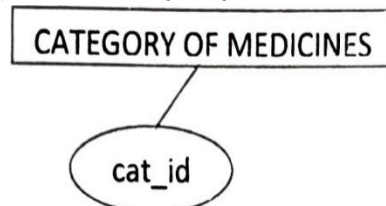
- As we have taken Order ID as the primary key in the entity "Bill Generation", the same attribute will be taken as an attribute in "Store Details" Entity as a foreign Key. Hence, we have established a foreign key in the Entity "Store Details".
- The "Store Details" Entity has all details related to the customers, orders and List of Medicines. Here, the "Store Details" Entity "HAS" the details about "List of Medicine". "HAS" is the relation formed between the two entities "Store Details" and "V of Medicines".



- M_ID is taken as a Primary Key in the Entity "List of Medicines" M_ID is the primary key as it uniquely identifies other attributes of the same entity.



- As we have taken M_ID as the primary key in the Entity "List of Medicines" the same attribute will be taken as an attribute in "Store Details". Hence, we have established a Foreign Key in the Entity "List of Medicines".
- There are different categories of Medicines and hence we have taken Category ID as a different Entity. In this we have taken cat_id (Category ID) as its Primary Key.



- Cat_id is taken as Foreign Key in the Entity "List of Medicines".
- Create a database based on above entities and attributes.

All Tables	▼	«	☐ Patient				
Patient	⌵		p_id	•	p_name	•	p_age
☐ Patient : Table			*		111 Hardik		17
Bill_Generation	⌵		*		222 Rufee		17
☐ Bill_Generation : Table			*		333 Zuha		17
Store_Details	⌵		*		444 Sohan		16
☐ Store_Details : Table			*				
List_of_medicines	⌵						
☐ List_of_medicines : Table							
Category_of_medicines	⌵						
☐ Category_of_medicines : Table							

All Tables	▼	«	☐ Bill_Generation				
Patient	⌵		order_id	•	p_id	•	price
☐ Patient : Table			*		121	111	200
Bill_Generation	⌵		*		122	222	300
☐ Bill_Generation : Table			*		123	333	400
Store_Details	⌵		*		124	444	100
☐ Store_Details : Table			*				
List_of_medicines	⌵						
☐ List_of_medicines : Table							
Category_of_medicines	⌵						
☐ Category_of_medicines : Table							

All Tables	Store_Details
Patient	store_id
Patient : Table	order_id
Bill_Generation	m_id
Bill_Generation : Table	date_of_issue
Store_Details	Add New Field
Store_Details : Table	
List_of_medicines	
List_of_medicines : Table	
Category_of_medicines	
Category_of_medicines : Table	

All Tables	List_of_medicines
Patient	m_id
Patient : Table	cat_id
Bill_Generation	m_name
Bill_Generation : Table	Add New Field
Store_Details	
Store_Details : Table	
List_of_medicines	
List_of_medicines : Table	
Category_of_medicines	
Category_of_medicines : Table	

All Tables	Category_of_medicines
Patient	cat_id
Patient : Table	cat_name
Bill_Generation	Add New Field
Bill_Generation : Table	
Store_Details	
Store_Details : Table	
List_of_medicines	
List_of_medicines : Table	
Category_of_medicines	
Category_of_medicines : Table	

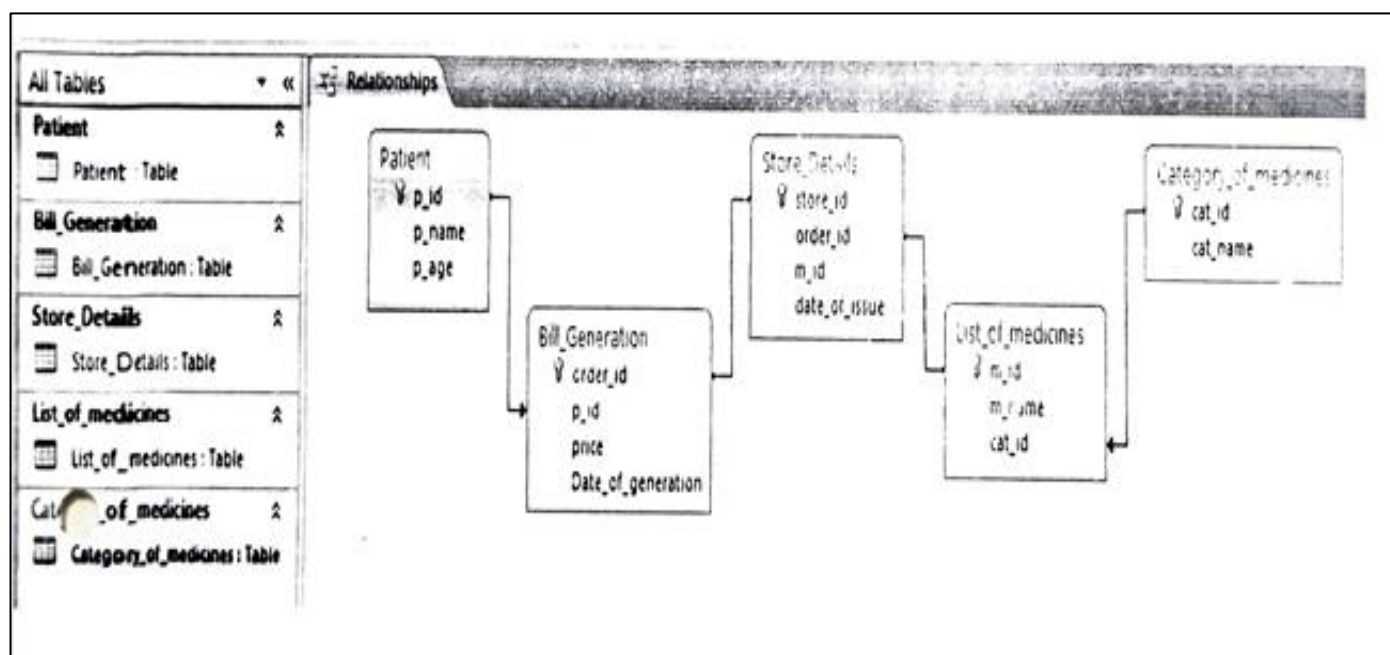
6. About The Project:

- The project is basically about making a database for medical billing system using important entities and attributes that are essential for making a simple database. The entities taken and the attributes are as follows:
- Patient: p_id (Primary Key attribute), p_name, p_age.
- Bill_Generation: p_id (Foreign key), order_id (Primary Key), price, Date_of_generation.
- Store_Details: store_id (Primary Key), date_of_issue, order_id (Foreign key), m_id (Foreign key).
- List of medicines: m_id (Primary Key), cat_id (Foreign key), m_name.
- Category_of_medicines: cat_id (Primary key), cat_name.
- We have taken all the essential details that are required to make a database and used it as entities.

7. Actual Resources Used:

Sr. No.	Name of Resources/Materials	Specification
1.	Microsoft Access 2010.	It provides users with one of the simplest and most flexible RDBMS solutions.
2.	Microsoft Word.	It is an automatically generated Microsoft Word document that serves as the functional specification component of your Axure RP file if needed,

8. Outputs of the Projects:



9. Skill Developed / Learning outcomes of this Project:

- Hence, we learnt to create a database system for medical billing system.
- Database is an effective and faster way of managing and retrieving information over a large scale.

10. Applications of this Project:

- This Database can be used to store and maintain details about different Customers and List of Medicines, Orders by the store.
- It can be used to access data about customer in a faster way.